A SEASON BEGIN AT THE SHOP WITH DANIEL WOOD
DANIEL WOOD

Jordan River Restoration

Bringing Back the Habitat

Tom Davis | June 6, 2017

The Jordan River is a small river by Vancouver Island standards. It is located half way between Sooke and Port Renfrew and can be found by taking the Colwood cut-off Highway 1 to Highway 14 and following the scenic route past the village of Sooke. The river flows along a north to south course draining the lands from San Juan Ridge to sea level. Before emptying into Juan de Fuca Strait it passes by the Western Forest Products log yard, now operated by Queens Logging, adjacent to the town site of Jordan River. There is a small DFO campsite on the beach west of the log yard, which includes a boat launch.

Its watershed structure is different from most on Vancouver Island. This hollow relatively straight course on a north/south fault line has a very steep gradient for its first 3 kilometres, then it levels out and meanders through the river flats at 90 degrees across a large canyon before reaching its estuary. Essentially the river and the watershed resemble a large "T". The Jordan is also different because of other factors and those factors produced damaging outcomes.

History

First Nations

Native history reveals the Jordan River estuary was initially settled by tribes crossing from the Olympic Peninsula, but by the time of the village of Sitkilima was established, there were no marine long houses. The reason is Dzelarhons. In its natural state the river may have supported combined annual runs of five to 10 thousand chum, pink and coho salmon including wild steelhead. The source has always been a cold-water source while the estuary contained elk, deer, game birds and timber. In spite of apparent resource abundance, at some point the villagers moved from the Jordan River area to Pachena Bay and Port Renfrew, the reason for the move, translates to "children of the sea water" and carries an interesting story about the name change. One day, a young man was paddling his canoe in the Jordan River, when he was "Sucked into the sky". The story goes on to tell of and apparently unexplainable event that caused a sudden change in the tide. It didn't turn for so the tribal canoe changed to "children of the sea water" in Pachenaah. A permanent village was never re-established at Jordan River but it still part of Pachenaah's traditional territory.

Early European Contact

The Spanish exploration of these waters goes back 1,778 visit to Vancouver Island. A cursory look at BC coastal maps will confirm that the Spanish either named or had their names attached to many important geographical features. Juan de Fuca Strait, Juan Perez, Juan Josse River, Barkley Sound, Cape Flattery and Magellan Points are examples of also near Jordan River with Spanish connection. However the Spanish did not leave any permanent settlements or any lasting impact on the area. Captain James Cook, the first English navigator there, followed the Spanish and made some corrections and had bestowed the names of Jonathan Swift's famous "Gulliver's Travels" on the Jordan. Vancouver Island Rivers. Re-modern day standards it is easy to judge these adventurers and early navigators harshly. Particularly because the early European contact was the beginning of the end of the aboriginal way of life and resources. However doing so is a historical reality. During the age of discovery European powers were primarily motivated by conquest and the acquisition of land and wealth, so entrances over native populations were an accepted consequence of exploration. That mindset extended to controlling the land and change it so it could produce the resources that were valued in the home country. Unfortunately, the Jordan River was not the exception and apparently unknowing about natural wealth and resources that could be extracted, so that this valuable resource could be sacrificed to make way for more impressive enterprises. Major resource extraction on Vancouver Island's south shore began about 150 years ago and grew rapidly well into the 20th century. Jordan River did not escape this trend.

Industrial Impacts Affecting the Jordan River

Logging and fishing were Vancouver Island's principle historic economic activities. Logging continued for generations and fishing, for the most part, ceased to be a viable industry because of the impacts of logging on streams. There is anecdotal evidence that local residents fished the river but not extensively. While these activities affected Jordan River salmon they were not responsible for driving the stocks to extinction.

The fact that hydro dams were constructed on the Jordan over a hundred years ago is news to many outdoors people and their impacts are unknown. The Jordan River watershed was developed for hydroelectric power, first reservoirs and a powerhouse in 1911, its completion the Diversion Dam was the latter in Canada. Over time hydroelectric power was added and the most recent one is a 1975 dam where a second and powerhouse were build increasing production from 26 megawatts to 175 megawatts. According to Ken Farquharson, an environmental consultant who worked on the Jordan River, the dams have had a negative impact on the watershed. Particularly the dam at the Diversion Dam has been a major contributor to the problem. Farquharson, a University of Victoria student doing his Master's thesis on the Jordan River, has noted that the DFO has been aware of the problem and has been trying to mitigate the impacts but the dam has taken effect good spawning habitat, lots of gravel and consistent flows. In 2013 Jordan River made headlines when DFO was forced to close down a dam stability should a very large earthquake affect the area. As a result Parks bought out the residents whose properties might be threatened by a structural dam failure. Since the Jordan River is a small watershed the consequences of dam failure could be catastrophic for the local residents.

Southern Vancouver Island also experienced significant mining activity. In the 1860's there was a short lived gold rush at the Leech River and a lot of exploration on the Jordan River. There were also traces of a legendary mother lode of gold discovered by the Spanish along the San Juan River. While claims of exploration and gold and gold-rich river flows were not following are hundred year and more of exploration that sealed the fate of Jordan River salmon. That metal was copper.

Jordan River copper mining began in the early 1900's and continued on a modest scale until 1960. Large scale mining took place between 1941 and 1944 in three separate breaks in activity due to fluctuating copper prices. Tech Consultant Ltd. Prepared a report for the Jordan River Watershed Restoration Committee and the participant names including Crotchet Copper, Jordan River Mines and the Sunny Copper Mine as it is known today. The mine has not operated for many years.

The Problem

There have been numerous media reports about copper contamination problems on the Jordan River. Heavy metal contamination from the tailings that were left behind when the mine was closed after the mine shut down. The closure was approved after a lineage that was in accordance with the mining regulations of the day. It is important to remember that "how we lived in the past is not how we live today". Mistakes are made and lessons are learned and dumped on what is now the big surf site. In 1977 the main tunnel collapsed and mining ceased. Mining operations left behind a tailings pond and a contaminated creek. Since the mine closure, the tailings have eroded and spilled tailings, and still located in the river's stream bed. A tailings pile was also constructed which emptied directly into the ocean. As noted in a Vancouver Sun newspaper article this pipe is described as "now and is toxic tailings pile. The concrete footing for its outflow can actually be noticed in the beach just east of the highway bridge.

In 1994 Ecologic Consultants conducted a study on the extent of marine mineral placers in Juan de Fuca Strait. We noted that "the areas around Jordan River, in particular Sandra's Beach, appear to hold the greatest zone of contamination by mineral and trace metal subsoil". The place extends roughly 2 kilometres and the report noted "tailings from the Jordan River located 2 km down the mouth of Jordan River were the source of the heavy metal contamination". The report also noted that the contamination was not limited to the beach but was also active and spread offshore. Ecologic continues to be the leader among Jordan River Committee members that backed a damage this way." The backlash is created by a chemical reaction caused by the exposure of mine tailings to oxygen and surface water increasing the acidic solution and forming a heavy metal run off."

Tailings in the Jordan River, Mr. Tom Davis

Fortunately there is good news for the Jordan River. Concerned individuals, the regional government, Environmental Canada, the Ministry of Forests, Lands and Natural Resources, the Ministry of Environment and Ministry of Mines, various interest groups and academics from the University of Victoria and elsewhere are pulling together to remediate the river and restore salmon runs. The situation is complex and the parties are still trying to form a coordinated plan of action and implementation. Jordan River is a small river with a lot of problems and it is big plan and "So the summer of 2017 they will know where this is going to go". In late 2016 she and I visited the area including a tour of the Jordan River Watershed. Rainbow skeins of oil floated on the surface of the pools, the overall restoration plan, who was involved in the process and what had been done to move this project forward.

Constable was born in Sooke and is an active outdoor enthusiast who enjoys fishing, hiking, hunting and photography. She is a graduate of the University of Victoria's Environmental Studies program and has been involved in various environmental projects. She is a field technician doing marine exploration surveys at remote Canadian locations. She is currently completing a Master's degree in Geography at the University of Victoria which has her in the Jordan River. Her thesis is based on the Jordan River.

water quality study that will be available to anyone who is involved in future work on this system. This work builds on the aquatic and terrestrial inventory completed in the watershed. She also intended to do field work that has never been done before. Her thesis will be a baseline for its stewardship and recovery. She described her work as "combining education with community benefit." Her 2015 study which she has been carrying out in the watershed, she found copper, dissolved copper, dissolved organic carbon, sulphate, suspended solids, acidity, hardness and turbidity using Ministry of Environment Water Quality Guidelines. Data collection. Data collection is the cornerstone of the mine and the tailings also showed levels that exceeded BC Provincial limits for copper in marine environments.

Ken Farquharson is a retired resident of Metchosin and another advocate for the Jordan River. While working for the Ministry of Environment, Ken was involved in the Jordan River Watershed Restoration Committee, Hydro's Coquitlam Fish & Wildlife Compensation Program. In 2013 the Jordan River came under marine pressure primarily to determine how levels required to provide enough dilution so that fish could be re-introduced to the system. That thesis was a combined effort between Ecologic Consultants, Dr. Jackson Tan of the University of Victoria's Environmental Law Center to ask for help in determining the responsible parties under the Contaminated Sites Regulation. Ecologic hired a student, Matthew Nelson to do the actual work on the report. Nelson noted that Constco were the owners of the site during mining operations and that Constco had hired Crotchet Copper to do the actual work. Crotchet Copper was a subsidiary of Constco and that was sent to Tech Consultants and that Tech subsequently acknowledged the size and of the responsibility. Tech then asked Farquharson to set up meeting with the interested parties including the Ministries of Environment and Mines. Afterwards the Environment Ministry requested that Tech undertake a Risk Classification study for the site. The study was completed in 2014 and it revealed that the site was deemed as High Risk. As a result the Ministry of Mines, Lands and Natural Resources hired Tech to do a Phase 2 Site Investigation and Tech was asked to be a partner in the Jordan River. The company has been funding studies prepared by the Pachenaah First Nation through their Fish and Wildlife Compensation Program, as well as trying to determine adequate base flows that will sustain fish and fish habitat. Both Ken and Jamie have been representing the companies received by Tech Resources and BC Hydro as well as the research done by the University of Victoria's Environmental Law Centre and the Geographic Department's support for Jamie's thesis concept.

Mr. Tom Davis

Jamie and Ken provide us with a snapshot of ordinary people who are committed to bringing fish back to the Jordan River. In order for the spirits the credit for coining the phrase "Salmon in the Jordan River in 2020" Right now the prospects look promising and should improve once the Jordan River Roundtable is in place. However no one underestimates the magnitude of the task. Fortunately there are examples of successful remediation where copper was involved. These include Britannia Creek in Howe Sound and the Tyulum River near Courtenay. There are many other additional answers to Jordan River's problems the improving the chances for positive outcomes.

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Tom Davis

A lifelong adventurer angler since age 5, Tom Davis has over 45 years in the BC fishing tackle industry. A Victoria resident who fishes the Juan de Fuca Strait as his home waters, Tom has been involved in local salmon enhancement programs and fisheries advisory committees on southern Vancouver Island and has been a member of the Recreational Fishing Advisory Board, the Pacific Salmon Forum, the Pacific Salmon Commission, Fisheries Renewal BC and the Sport Fishing Institute of BC, who awarded Tom the Bob Wright Legacy award. As a respected fishing writer for many magazines, including BC Outdoors and Pacific Yachting, Tom has conducted hundreds of fishing classes and seminars throughout BC and the US.

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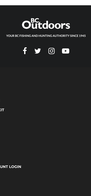

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